Multimodal Fusion of 3D CT Foundation Model Embeddings and Clinical Data for Enhanced Prognosis of Acute Exacerbations in COPD

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[1] Anonymized Affiliations
email@anonymized.com

Abstract. Chronic obstructive pulmonary disease (COPD) is the third leading cause of death globally, and acute exacerbations (AECOPD) of COPD are key drivers of morbidity. Current clinical trial enrollment strategies for COPD rely heavily on airflow limitation and exacerbation history, which often fail to capture the structural heterogeneity of fast-progressing phenotypes. To improve disease prognosis, we propose utilizing a multimodal framework that leverages high-resolution computed tomography (HRCT) images containing critical prognostic information absent from standard clinical metrics in COPD patients. Our approach utilizes Merlin, a 3D CT foundation model primarily pretrained on diverse abdominal scans rather than lung-specific data. Despite this cross-domain pretraining, we show that the model extracts rich latent representations capable of predicting COPD progression. To integrate these high-dimensional image embeddings with a smaller set of clinical data and quantitative Computed Tomography (qCT) metrics selected using the XGBoost method, we employed an encoder-based fusion strategy to align modalities into a balanced joint representation space and trained a ML model on the combined resulting feature set for prognosis of acute exacerbations. Validated on the COPDGene dataset, our multimodal model demonstrated improved performance for prognosis of AECOPD (AUC=0.87) over clinical metrics (AUC=0.76) and image-based (AUC=0.66) unimodal ML models. These results further validate the use of HRCT as a prognostic modality and demonstrate that medical foundation models can effectively unlock latent information within HRCT data that improve identification of fast-progressing patients over current gold-standard clinical markers. This work provides a foundation for more efficient clinical trial enrichment through image-based prognosis, pending further validation in prospective clinical trials.

Keywords: Multimodal AI, COPD Exacerbation Prognosis, CT Foundation Model.

1 Introduction

The global burden of chronic obstructive pulmonary disease (COPD) - a type of obstructive lung disease characterized by persistent airflow limitation - has reached a critical juncture, characterized by high mortality rates - third leading cause of death worldwide- and an escalating economic impact that strains modern healthcare systems. Central to the disease's morbidity are acute exacerbations (AECOPD) -sudden and

sustained worsening of respiratory symptoms such as dyspnea, cough, and sputum production - that drive lung function decline, hospitalizations, and mortality. Identification of COPD patients at risk for these events is of high value for advancing the care of patients with COPD and drug development [1,2]. [1,2].

COPD clinical trial enrichment strategies have long relied on spirometric measurements, specifically the forced expiratory volume in one second (*FEV1*), forced vital capacity (*FVC*), or history of exacerbations, to enroll patients at risk of progression. However, *FEV1* reflects only a narrow segment of the physiological changes occurring within the lung and often does not robustly identify individuals at high risk for exacerbations.This "prognostic gap" has underscored the need for the development of more sophisticated artificial intelligence (AI) frameworks capable of integrating multimodal data—including high-resolution computed tomography (CT) imaging, longitudinal electronic health records (EHR), blood-based biomarkers, and even acoustic signals—to provide a comprehensive and dynamic assessment of patient risk [3–5].

AI in pulmonary medicine has shifted from task-specific algorithms to multimodal foundation models (FMs) that mimic clinical decision-making. Unlike traditional AI optimized for single tasks like nodule detection, FMs are large-scale architectures that provide rich prior knowledge, allowing for diagnosis and risk stratification with minimal adjustment [5–7]. This is vital for COPD, where patients with similar lung function *(FEV1*) often exhibit vastly different structural pathologies [8,9]. Furthermore, because documented exacerbation events in cohorts like COPDGene [10] are limited, the use of FMs as feature extractors is necessary to overcome the scarcity of labeled data for training models from scratch.

Quantitative Computed Tomography as a Prognostic Pillar

Computed tomography (CT) serves as the primary modality for visualizing the structural manifestations of COPD. Quantitative Computed Tomography (qCT) has emerged as a critical prognostic pillar, offering a non-invasive window into the spatial heterogeneity of emphysema, air trapping, and airway wall thickening. HRCT-derived data can distinguish between phenotypes that appear clinically similar but follow vastly different trajectories of disease progression [8, 9, 11, 12]. In this paper, we explore the hypothesis that latent features extracted from HRCT through medical image foundation models contain high-fidelity prognostic signals in addition to qCT based features that are currently underutilized in COPD clinical trials.

To demonstrate this, we developed a benchmarking framework using the COPDGene dataset. We first established a baseline performance using a "Clinical-Only" machine learning model comprising 12 key tabular features, including patient demographics, smoking history, and quantitative Computed Tomography (qCT) metrics. We then compared this against an "HRCT-Only" AI approach utilizing high-dimensional embeddings from Merlin, a 3D CT foundation model. Lastly, we introduced multi-modal frameworks that integrated complementary clinical and imaging embeddings to improve prediction of future COPD exacerbations. We

systematically evaluated proposed frameworks against unimodal baselines to quantify the added prognostic value of multimodal integration in COPDGene.

Summary of Contributions

In this work, we present a multimodal framework that leverages high-resolution computed tomography (HRCT) scans and clinical data to identify fast-progressing COPD patients at risk of future acute exacerbations. Our approach integrates latent imaging features extracted from medical FMs with structured clinical variables, and is, to our knowledge, the first to systematically benchmark FM-derived imaging embeddings against unimodal baselines for acute COPD exacerbation prediction. We rigorously evaluated the framework across multiple machine learning architectures and validated on the longitudinal *COPDGene* cohort, demonstrating consistent performance gains over unimodal models. To enhance interpretability, we implemented an Extreme Gradient Boosting (XGBoost) Gini importance pipeline to identify key clinical and quantitative CT (qCT) predictors. Our main contributions are three-fold:

1. A multimodal framework that integrates HRCT-derived latent imaging features with clinical data to improve prediction of acute COPD exacerbations.
2. Extensive evaluation of state-of-the-art medical imaging foundation models, including Merlin, MedSAM2, and CT-CLIP, demonstrating that Merlin—despite being pretrained primarily on abdominal and whole-body CT volumes—captures pulmonary prognostic signals relevant to exacerbation risk.
3. Implementation of XGBoost Gini importance pipeline to identify key clinical and qCT features

2 Methodology

Our approach follows a hierarchical validation strategy: establishing a robust clinical baseline, extracting high-fidelity latent representations from HRCT using foundation models, and fusing these modalities through a dimensionality-balancing architecture. Finally, multiple machine learning models are used on the combined feature set for prediction of future exacerbations. Figure 1 shows the model architecture.

2.1 Clinical Benchmark Model and Feature Selection

We utilized tabular features from the COPDGene dataset, including demographics, smoking history, and quantitative imaging endpoints. We employed an XGBoost classifier, favored for its ability to handle non-linear relationships in clinical data. Feature selection was driven by Gini Importance (Gain) to ensure the model utilized only the most predictive clinical signals.

Analysis of the XGBoost model identified Exacerbation History as the most significant predictor of future events, followed by age and physiological metrics. To validate the clinical soundness of our benchmark, we utilized SHAP (SHapley Additive

exPlanations) to visualize feature contributions. The SHAP summary plot reveals that higher values of %Emphysema and a significant history of prior exacerbations are strongly associated with a higher predicted risk of future acute exacerbations. The resulting benchmark model achieved a mean AUC of 0.76±0.08 . While this represents a strong clinical baseline, the reliance on historical events and coarse physiological markers leaves a "prognostic gap" that we hypothesize can be improved by latent imaging features.

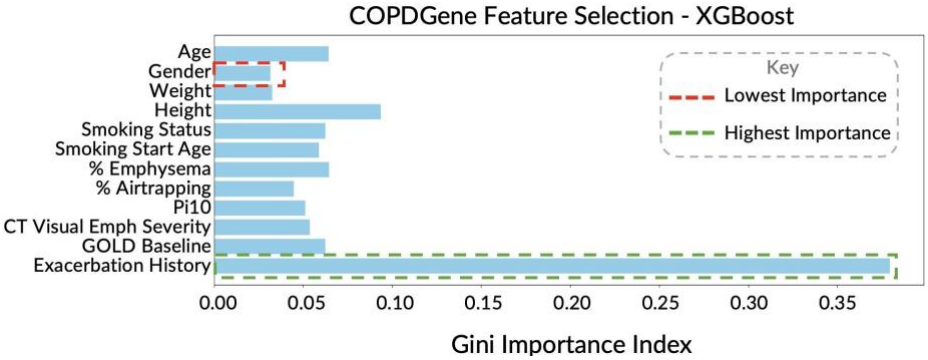

Fig. 1. Gini importance of tabular features (XGBoost). Exacerbation history was the most important feature, while gender was the least

2.2 Foundation Models for HRCT Embedding Extraction

We investigated three state-of-the-art medical image foundation models (FMs) to determine their efficacy in capturing prognostic pulmonary signals: Merlin, MedSAM2, and CT-CLIP [13–15].CT-CLIP is a contrastive language-image model pretrained on non-contrast 3D chest CT scans to align volumetric data with radiology text reports. MedSAM2 is a more versatile model pretrained on 78 diverse medical datasets across multiple modalities (HRCT, MRI, X-ray) for robust segmentation and feature extraction. Lastly we explored Merlin, a 3D CT foundation model with an 3D ResNet152 backbone, pretrained on a large internal dataset of CT scans from 18,321 patients.

Preliminary evaluations across various downstream tasks, including COPD disease severity classification, demonstrated that Merlin provided the most meaningful and discriminative features for COPD patients (Gold 0 vs. Gold 4 AUC: 0.97). Consequently, our multimodal framework focuses on the extraction of latent representations specifically from the Merlin vision backbone.

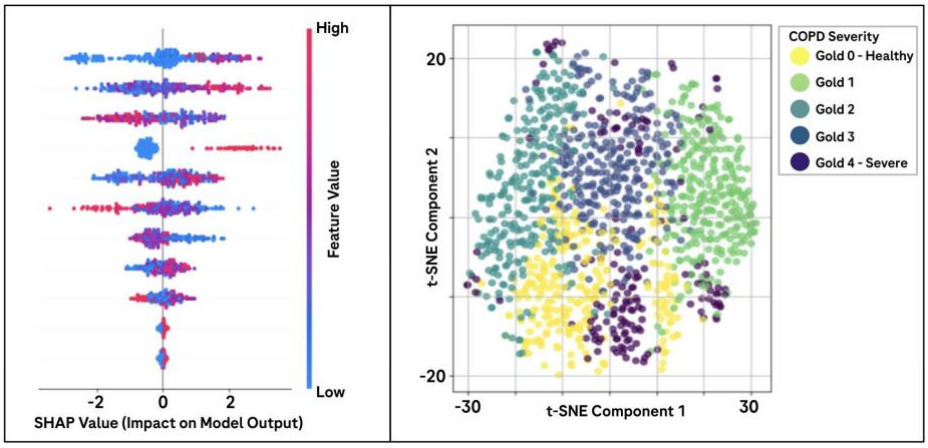

Fig. 2. Left. SHAP-based feature contributions for COPD exacerbation predictions — positive (red) and negative (blue) Right. t-SNE visualization of Merlin embeddings from COPDGene CT scans, colored by final GOLD scores

2.3 Preprocessing and Embedding Pipeline

To ensure consistency across the COPDGene, we implemented a standardized preprocessing and extraction workflow. For the Merlin encoder, volumetric data was resized and resampled to a fixed dimension of $512\ x\ 512\ x\ 224$ (H x W x D). We performed Hounsfield Unit (HU) windowing to focus on relevant tissue densities (windowing ranges included (-600, 1500) and (-1000, 0) followed by normalization to a range of [0, 1].

Embedding Extraction and Storage The preprocessed HRCT volumes were passed through the frozen Merlin 3D image encoder. This process projected the high-resolution volumetric data into a high-dimensional latent space, resulting in a feature vector of size 2048 per scan. To optimize computational efficiency during downstream model training and cross-validation, these embeddings were extracted and saved in a structured database, serving as fixed visual descriptors for each patient.

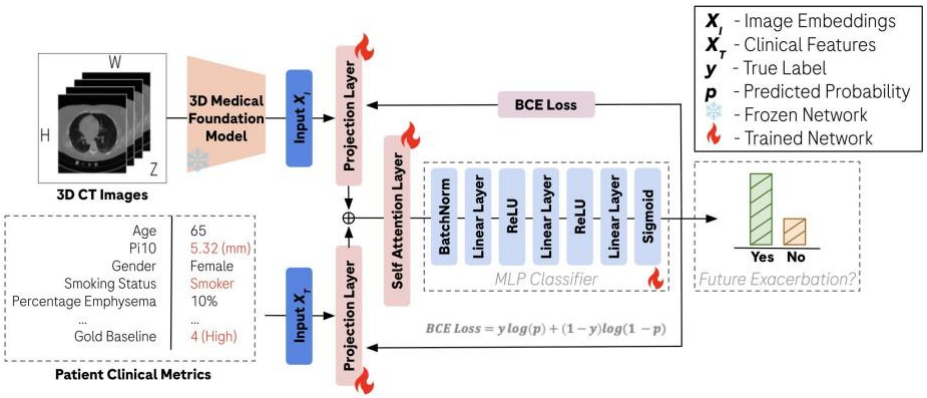

Fig. 3. Proposed multimodal architecture for exacerbation risk prediction in COPD patients. HRCT image embeddings X_I (extracted using FM) and clinic features, X_T are each projected into 640-dimensional latent spaces and concatenated to form 1280dimensional fused representations. The fused embedding is passed to a MLP classifier comprising batch normalization, two 10-unit fully connected layers with ReLU activations, and a final linear layer with sigmoid activation for binary prediction.

2.4 Dimensionality-Balanced Multimodal Fusion

A major challenge in multimodal COPD prognosis is the mismatch between high-dimensional imaging data (2048 dimensions) and a small set of clinical features (12 features). To resolve this, our fusion architecture uses independent fully-connected layers to project both modalities into a joint latent space. Hyperparameter optimization showed that compressing visual embeddings into a 640-dimensional vector achieved the best balance of feature richness and stability. This balanced vector is then processed by a multi-layer perceptron (MLP) with two hidden layers (size 10), using Batch Normalization, ReLU, and dropout to prevent overfitting on the limited positive samples. See Figure 1.

3 Experiments and Results

Dataset: To train and validate our model, we utilized the COPDGene study (Genetic epidemiology of COPD) study [10]- one of the largest studies to document the unique patterns of progression of COPD, consisting of HRCT and corresponding clinical data.The dataset includes approximately 10,000 participants at baseline and roughly 5,000 participants at the follow-up stage at 5 year interval. For model development, we employed 5-fold cross-validation specifically on the subset of patients from COPDGene with available exacerbation data, totaling approximately 458 samples**.** We balanced the data with randomly selecting patients with no exacerbations recorded from the rest of the data to train a binary classifier**.** These samples were evenly distributed across multiple folds for training and validation of the model.

We evaluated image embeddings from state-of-the-art medical Foundation Models and visualized the high-dimensional image embeddings extracted by projecting into a 2D space using t-SNE (t-Distributed Stochastic Neighbor Embedding) across four different perplexity (P) settings (10, 30, 50, and 80).The distinct separation between the purple and yellow clusters suggests that the foundation model embeddings have successfully captured features in the CT scans that correlate with severe exacerbation risk(see Fig. 4). These results suggest that foundation models can "see" pathological differences in lung tissue without needing explicit disease labels during the feature extraction phase.

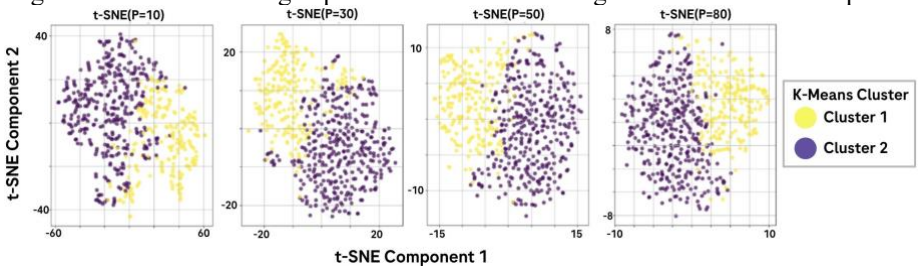

Fig. 4. As perplexity, P, increases from P=10 to P=80, the clusters become more defined and global structures are better preserved. At P=80, the clear bifurcation indicates that the model is identifying two primary "latent phenotypes" within the severe exacerbation class

Simple concatenation of foundation model embeddings and clinical metrics degrades performance; however, our proposed fusion strategy significantly outperforms unimodal models. The Merlin-based multimodal framework achieved the highest performance with an AUC of 0.87±0.04, vastly exceeding the history-of-exacerbation baseline (AUC=0.58) currently used in clinical trial enrollment. Other configurations, including MedSAM2 and CT-CLIP, also showed strong results (AUC=0.87 and 0.84, respectively). Subgroup analysis revealed minimal gender bias, with False Negative Rates (FNR) of 0.27 for females and 0.30 for males.

To our knowledge there is no other study on COPDGene for prediction of acute exacerbations that uses AI extracted image embeddings in particular by utilizing medical image Foundation models to build a multimodal model and benchmark against unimodal clinical metrics or HRCT based models. The closest work is the study performed by Chaudhuri et al., 2023 [16] and SPIROMICS [17] which found combining quantitative CT (airway structure and density texture) with clinical history significantly outperformed the standard BODE index (AUC = 0.812) for predicting severe exacerbations over a 3-year period.

Table 1. Performance of multimodal AI model against each unimodal model

Model	AUC	Accuracy	F1 score	Precision	Recall

Prior Exacerbation	0.58	0.58	0.35	0.75	0.23
Clinical Only Model (A)	0.76±0.08	0.67±0.08	0.66±0.08	0.69±0.09	0.64±0.09
Imaging Only Model (B)	0.66±0.09	0.64±0.08	0.63±0.09	0.66±0.11	0.61±0.12
Multimodal Model (A+B)	**0.87±0.04**	**0.78±0.07**	**0.78±0.04**	**0.78±0.11**	**0.81±0.08**

4 Conclusions

The results of this study underscore a significant "prognostic gap" in current COPD clinical trial enrichment strategies. While traditional metrics like FEV1 and exacerbation history offer a snapshot of functional status, they do not capture the localized structural damage that precedes clinical decline. Our analysis confirms that history of severe/moderate exacerbations alone is a relatively weak predictor of future events, even though a history of exacerbation is the best clinical predictor of future exacerbation. By utilizing Merlin, a 3D CT foundation model, we demonstrated that cross-domain pretraining, even on non-lung-specific abdominal data, can extract high-fidelity latent representations that correlate strongly with COPD progression. The use of t-SNE visualization further validated this, revealing clear "latent phenotypes" within the severe exacerbation class that are visible to the model without explicit disease labels.

A critical technical hurdle addressed in this work was the dimensional mismatch between the 2048-dimensional visual embeddings and the 12-feature clinical vector. Our specialized dimensionality-balanced fusion architecture successfully prevented "modality drowning". By projecting visual embeddings to a 640-dimensional space and integrating them with clinical data, the multimodal model achieved a superior AUC of 0.87, significantly outperforming the Clinical-Only (AUC=0.76) and Image-Only (AUC=0.66) benchmarks.This research provides a robust foundation for enhancing COPD prognosis through the integration of high-resolution computed tomography (HRCT) and clinical data. We have shown that medical foundation models effectively unlock latent prognostic information within HRCT scans that standard clinical markers miss.

For the successful translation and deployment of this model, a critical next step involves further adjustment and training with more granular longitudinal datasets. Specifically, future work will focus on utilizing datasets such as SPIROMICS, which collects exacerbation data more frequently (annually) than the 5-year intervals found in the current cohort. Training on such data will allow for more precise temporal modeling of disease progression. Ultimately, this refined model must be validated on prospective clinical trial data to confirm its utility as a tool for clinical trial enrichment and the identification of fast-progressing patients.

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